Tutorial 3 20/1/26

A mobile service provider wants to classify SMS messages as Spam or Not Spam using a single-layer neural network in the layer. with 5 ~~fro~~ neurons in the ~~hidden~~ hidden layer.

x_1 = specific words and phrases

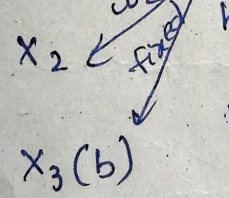
x_2 = Links and Attachments

x_3 = Bias (const) (doesn't depend on anything)

x_1	x_2	x_3 (b)	Target(y)
4	2	1	1
1	0	1	0
6	3	1	1
2	1	1	0

Hidden layer

Inputs H.L Output



$v_1 = 0.1$

$v_2 = 0.2$ 0 or 1

$v_3 = 0.3$

$v_4 = 0.4$

$v_5 = -0.3$

(hidden layer & output → weight)

(input & hl)

	w_1	w_2	b
h_1	0.1	0.2	$1 \to z_1$
h_2	-0.2	0.3	$1 \to z_2$
h_3	0.4	-0.1	$1 \to z_3$
h_4	0.3	0.2	$1 \to z_4$
h_5	0.2	-0.4	$1 \to z_5$

$z_1 = ?$ h_1 (only one input)

$x_1 = 4 \quad x_2 = 2 \quad , b = 1$

$w_1 = 0.1 \quad w_2 = 0.2$

$z_1 = 0.4 + 0.4 + 1$

$\boxed{z_1 = 1.8}$ $\boxed{\hat{y} = 1}$

$z_4 = ?$ h_4

$x_1 = 4 \quad x_2 = 2 \quad b = 1$

$w_1 = 0.3 \quad w_2 = 0.2$

$z_4 = \cancel{0.6 + 0.2 + 1}$

$\boxed{z_4 = 2.6}$ $\boxed{\hat{y} = 1}$

h_2

$z_2 = ? \quad x_1 = 4 \quad x_2 = 2 \quad , b = 1$

$w_1 = -0.2 \quad w_2 = 0.3$

$z_2 = \cancel{-0.2 + 0 + 1}$

$\boxed{z_2 = 0.8}$ $\boxed{\hat{y} = 1}$

h_5

$z_5 = ?$

$x_1 = 4 \quad x_2 = 2 \quad , b = 1$

$w_1 = 0.2 \quad w_2 = -0.4$

$\boxed{z_5 = 1}$ $\boxed{\hat{y} = 1}$

$\Rightarrow w_1 x_1 + w_2 x_2 + b$

$z_3 = ?$ h_3

$x_1 = 4 \quad x_2 = 2 \quad b = 1$

$w_1 = 0.4 \quad w_2 = -0.1$

$z_3 = \cancel{2.4 + 0.3 + 1}$

$\boxed{z_3 = 2.4}$ $\boxed{\hat{y} = 1}$

$$h_1 = \sigma(z_1) = \frac{1}{(1+e^{-z})} \text{ (sigmoid fn)}$$

$$= \frac{1}{1+e^{-1.8}} = 0.86$$

$$\sigma(z_2) = \frac{1}{1+e^{-0.8}} = 0.69$$

$$\sigma(z_3) = \frac{1}{1+e^{-2.4}} = 0.92$$

$$\sigma(z_4) = \frac{1}{1+e^{-2.6}} = 0.93$$

$$\sigma(z_5) = \frac{1}{1+e^{-1}} = 0.73$$

Final: $z_{output} = h_1 v_1 + h_2 v_2 + h_3 v_3 + h_4 v_4 + h_5 v_5 = ?$

$$= (0.86)(0.1) + (0.69)(0.2) + (0.92)(0.3) + (0.93)(0.4) + (0.73)(-0.3)$$

$$= 0.377 \approx 0.38$$

$$\sigma(z_{out}) = \frac{1}{(1+e^{-0.38})} = 0.59 \approx 0.60$$